# Spot Micro - final parts list

31 Oct 2026. Replaces the 23 Oct list. Checked against the KDY0523 _cls frame in the 3D model. 3D-printed parts: see print/README.md.

## 1. Computing & sensors

| | Item | Qty | Notes |
|---|---|---|---|
| ☐ | Raspberry Pi 3B | 1 | |
| ☐ | microSD card, 16-32 GB | 1 | |
| ☐ | Raspberry Pi Camera v2 | 1 | |
| ☐ | Camera ribbon cable, 15-pin | 1 | 30 cm (route is ~22 cm) |
| ☐ | MPU-6050 (IMU) | 1 | |
| ☐ | LCD 16x2 with I2C backpack | 1 | Address 0x27 |
| ☐ | I2C level shifter, BSS138, 4-ch | 1 | LCD backpack pulls SDA/SCL toward 5 V; Pi GPIO is 3.3 V only |
| ☐ | Heatsink for Pi CPU | 1 | Skip if already installed |
| ☐ | Dupont jumper wires F-F | 1 set | |

## 2. Drive

| | Item | Qty | Notes |
|---|---|---|---|
| ☐ | PCA9685 servo driver, 16-ch | 1 | |
| ☐ | Servo PDI-HV5523MG | 12 | |
| ☐ | Servo extension 3-pin, 20 cm | 4 | Knee servos |
| ☐ | Servo extension 3-pin, 10 cm | 2 | Right hip servos. Other 6 servos: own cable is enough |

## 3. Power

| | Item | Qty | Notes |
|---|---|---|---|
| ☐ | LiPo 2S 4000-5000 mAh, shorty hardcase | 1 | Must fit bay: max 127 x 48 x 25.5 mm (check height and width). Standard 138 mm packs do not fit. E.g. Gens Ace 2S 5000 mAh shorty, 95 x 46.8 x 25 mm |
| ☐ | UBEC 5 V / 5 A, 2S input | 1 | E.g. Hobbywing UBEC 5A (2-8S, 5 A cont.). Set output to 5.0 V before connecting the Pi |
| ☐ | Rocker switch Switch Components RF1-1A-DC-2-B | 1 | 16 A at 12 V DC, panel cut-out 19.2 x 13 mm. 20 A rockers do not fit |
| ☐ | Blade fuse 15 A (ATC) + inline holder, 14 AWG | 1 | Not 20 A: switch is rated 16 A |
| ☐ | XT60 connector pair | 1 | Solder to the battery if it comes with bullets or T-plug |
| ☐ | XT60 Y-splitter | 1 | Servos + UBEC |
| ☐ | Capacitor 1000-2200 uF, 16 V or higher | 1 | On servo power line |
| ☐ | Silicone wire 14-16 AWG, red + black | 1 m each | |
| ☐ | 2S LiPo balance charger | 1 | |
| ☐ | LiPo low-voltage buzzer | 1 | |

## 4. Voice

| | Item | Qty | Notes |
|---|---|---|---|
| ☐ | USB microphone | 1 | |
| ☐ | USB speaker | 1 | |

## 5. Optional: battery voltage

| | Item | Qty | Notes |
|---|---|---|---|
| ☐ | ADS1115 ADC | 1 | |
| ☐ | Resistors 20 kOhm + 10 kOhm | 1 + 1 | Divider: 8.4 V -> 2.8 V |

## 6. Fasteners & bearings

| | Item | Qty | Notes |
|---|---|---|---|
| ☐ | Bolt M5x15 | 8 | |
| ☐ | Bolt M4x20 | 40 | |
| ☐ | Bolt M4x15 | 8 | |
| ☐ | Nut M4 | 48 | |
| ☐ | Bolt M3x20 | 4 | |
| ☐ | Bolt M3x10 | 36 | Frame 28 + shoulder reinforcement 8 |
| ☐ | Nut M3 | 24 | Frame 16 + shoulder reinforcement 8 |
| ☐ | Bearing F625zz | 12 | Frame 8 + shoulder reinforcement 4 |
| ☐ | Screw M2x6 + nut M2 | 4 + 4 | Camera mount |

## 7. Consumables

| | Item | Qty | Notes |
|---|---|---|---|
| ☐ | Self-tapping screw 2 x 6-8 mm | 8 | Pi (4) and PCA9685 (4) to their printed platforms (holes 1.5 mm) |
| ☐ | Double-sided foam tape | 1 roll | Pi and PCA9685 platforms to plate; MPU-6050 to plate |
| ☐ | Hot glue sticks + glue gun | 1 | Servo horns into printed parts (KDY0523 instructions) |
| ☐ | Zip ties, small | 1 pack | Servo cables |

# Wiring

Same as the mike4192 build (2S direct to the PCA9685 servo terminal, UBEC 5 V for the Pi). Differences: PCA9685 logic at 3.3 V and a level shifter for the 5 V LCD, so the Pi I2C pins never see 5 V. The mike4192 software drives only the PCA9685 and the LCD; MPU-6050, camera, ADS1115 and voice need our own code.

## Power (14-16 AWG)

| From | To | Notes |
|---|---|---|
| LiPo 2S | XT60 | Battery plug |
| XT60 (+) | Fuse 15 A -> switch -> Y-splitter (+) | Switch in the rear cover |
| XT60 (-) | Y-splitter (-) | |
| Y-splitter branch 1 | PCA9685 V+ / GND terminal | Servo power 6.4-8.4 V; capacitor across this terminal |
| Y-splitter branch 2 | UBEC input | |
| LiPo balance lead | Low-voltage buzzer | |

## 5 V (UBEC output)

| From | To | Notes |
|---|---|---|
| UBEC 5 V / GND | Raspberry Pi pin 2, 4 (5 V) / pin 6 (GND) | Output jumper at 5.0 V |
| UBEC 5 V / GND | LCD backpack VCC / GND, level shifter HV | |

## I2C bus, 3.3 V (Dupont)

| From | To | Notes |
|---|---|---|
| Pi pin 1 (3.3 V) | PCA9685 VCC, MPU-6050 VCC, ADS1115 VDD, level shifter LV | PCA9685 logic at 3.3 V, not 5 V |
| Pi pin 3 SDA / pin 5 SCL | PCA9685 (0x40), MPU-6050 (0x68), ADS1115 (0x48) | No address conflicts |
| Pi pin 3 / 5 via level shifter | LCD SDA / SCL (0x27) | |
| Pi pin 9 (GND) | PCA9685 GND, all module GND | Common ground |

## Signals

| From | To | Notes |
|---|---|---|
| PCA9685 channels | 12 servos (3-pin plugs on the header) | Knee servos via 20 cm, right hips via 10 cm extensions |
| Battery (+) after switch | Divider 20k/10k -> ADS1115 A0 | Optional, max 2.8 V at A0 |
| Camera v2 | Pi CSI connector, 30 cm ribbon | |
| USB mic, USB speaker | Pi USB ports | |